PERFORMANCE WORK STATEMENT

SAMM Engineering Deployment and Support - NOAA

POP: Date of award through 5 years thereafter

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1. BACKGROUND:

Standardized use of a Centralized Maintenance Management System (CMMS) is mandated under International Maritime Organization (IMO) requirements, as implemented through ABS, SOLAS, and MARPOL conventions. To meet these obligations, NOAA utilizes a program called Shipboard Automated Maintenance Management (SAMM) to plan, document, and verify completion of maintenance and repair activities across the fleet.
SAMM was originally developed for the Military Sealift Command (MSC) and has been in use across the NOAA fleet for over 10 years. The system complies with federal IT and data management requirements and has been integrated into both shipboard and shoreside operations. It has proven effective in supporting NOAA's mission by improving lifecycle management, optimizing preventative maintenance, and ensuring compliance with maritime regulations.

The system includes multiple integrated modules such as Inventory Spare Parts Management, Vibration Monitoring System (VMS), Oil Analysis System (OAS), the Electronic Engineering Logbook, and the Electronic Vessel Logbook (ShipsLog). These tools allow NOAA to track asset condition, streamline logistics, and perform advanced condition monitoring. Technical assistance under this contract will include database development and maintenance for each ship, shipboard software installation and updates, software and data upgrades, analysis of vibration and oil sample data, and ongoing training and user support.

The previous SAMM IDIQ contract has reached the end of its period of performance. NOAA has made substantial investments over the past decade in SAMM deployment, system configuration, and workforce training. Disruption to the continuity of SAMM support would create significant risk to maintenance tracking, regulatory compliance, and mission readiness.

Awarding a follow-on contract ensures uninterrupted support of the system and leverages NOAA's existing investment in SAMM. It also preserves data continuity and system configuration unique to NOAA's fleet. NOAA intends to continue use of SAMM as the agency-wide standard for shipboard maintenance management. This follow-on IDIQ contract is essential to maintain operational capability, regulatory compliance, and the long-term value of the SAMM program.

2. SCOPE:

Provide engineering services to the NOAA fleet to maintain, develop and update the Shipboard Automated Maintenance Management (SAMM) system, the SAMM Spare Parts Module, the Ship Inspection Program (SIP), the Vibration Monitoring System, the Oil Analysis System, future SAMM Addons, Virtual Technical Library (VTL), the Logbook and Ship's Log, training to afloat personnel, training to ashore personnel, and training NOAA officers in the use of these systems. Provide SAMM support during system upgrades.

Vessel Support: This task is for supporting NOAA Vessels currently in operation (15) and under construction (2), future vessels in planning (4), and decommissioning of current vessels (5), including but not limited to:

i. *Bell M. Shimada*
ii. *Fairweather*
iii. *Ferdinand R. Hassler*
iv. *Gordon Gunter*
v. *Henry B. Bigelow*
vi. *Nancy Foster*
vii. *Okeanos Explorer*
viii. *Oregon II*
ix. *Oscar Dyson*
x. *Oscar Elton Sette*
xi. *Pisces*
xii. *Rainier*
xiii. *Reuben Lasker*
xiv. *Ronald H. Brown*
xv. *Thomas Jefferson*
xvi. *Oceanographer*
xvii. *Discoverer*
xviii. *Future Vessels (x4)*

3 SPECIFIC TASKS:

3.1 Engineering Support:

3.1.1 *Operational Shipboard Systems Support*. The Contractor shall visit NOAA ships on an as needed basis to provide the following:

3.1.1.1 Support SAMM systems aboard ships to include installation of the latest approved shipboard version of SAMM with the intent to remain aligned with Military Sealift Command (MSC), reinstallation, training, and troubleshooting support for SAMM software and related data collectors. Provide updates to NOAA Afloat SAMM Versions within 180 days of release to MSC of a particular version of SAMM unless the update includes a Cybersecurity Remediation. Cybersecurity remediations shall be delivered within 30 days of delivery to MSC.

3.1.1.2 Input complete and missing equipment maintenance actions into the SAMM system while onboard and train the ship's crew to maintain the data using the SAMM system.

3.1.1.3 Meet with the ship's crew to determine billet-to-PM (Preventative Maintenance) code requirements, document in the Monthly Summary Report, and input data identifying billet-to-PM code requirements into the SAMM software.

3.1.1.4 Perform and provide on-the-job-training (OJT) in the collection of condition monitoring samples.

3.1.1.5 Provide onboard troubleshooting and OJT of Logbook/ShipsLog and associated cross-system interfaces.

3.1.1.6 Provide onboard investigative work to resolve help calls and feedback requests.

3.1.1.7 Generate a complete and up-to-date planned maintenance schedule in SAMM for the fleet as specified above. The schedule will be provided to the ship's crew for verification. The schedule shall reflect the current status of all maintenance actions. Distribute the ships' planned maintenance workload evenly throughout the year.

3.1.1.8 Perform a baseline inventory of spare parts and insert/update the data in the NOAA Spare Parts Module. Provide training to the ship's crew to maintain the data.

3.1.1.9 Provide onboard OJT for the ship's crew for the use of SAMM. The engineering department, Department Heads and other designees shall be trained on the use and contents of the SAMM system manuals that are provided with each SAMM installation.

3.1.1.10 Provide technical assistance 24 hours a day, 7 days a week via email, phone, and on-site visit if not resolved remotely for data collectors and hardware, software and networking installation or troubleshooting as it pertains to the SAMM system. Anticipated travel for this support is included in the Table below.

3.1.1.11 Build and Maintain the Logbook and ShipsLog Module data on ships including survey and verify machinery gauges and add and update equipment data (estimated 15 ships).

3.1.1.12 Provide initial and refresher remote SAMM training courses for Afloat Personnel. These courses should be targeted to the knowledge level of the individual users. The NOAA fleet comprises approximately 200 individuals who will interact with SAMM at various levels.

3.1.1.13 For estimation purposes the Contractor shall estimate for engineer or technician support as required to perform the number of trips associated with each of the locations listed below each year. The Contractor shall coordinate with the NOAA TPOC and the COR targeted for support. It is estimated that each trip will include multiple ship visits at the locations specified with an estimated total duration per trip of 3 days onsite related to all Operational Shipboard Systems (SAMM) Support & 5 days for all SAMM Software Upgrade Support visits such as ship checks etc. Please see the estimated travel table below.

Table1. Yearly Estimated Travel Table

Location	Number of trips	Personnel Supporting	Days On-site	Travel Days
SAMM Engineering Support & Training Visit				
Charleston, SC	2	2	3	2
Pascagoula, MS	3	2	3	2
Honolulu, HI	2	2	3	2
Newport, OR	2	2	3	2
New Castle, NH	1	2	3	2
West Coast, CONUS	3	2	3	2

Ketchikan, AK	1	2	4	2
SAMM Software Upgrade Support visit				
Charleston, SC	1	2	5	2
Pascagoula, MS	3	2	5	2
Honolulu, HI	1	2	5	3
Newport, OR	2	2	5	2
San Diego, CA	1	2	5	2
Monthly Meeting Support & Local Travel				
Location	Round Trip Trips			
Within 50 miles of Norfolk, VA and Newport, RI area vessels	12		2	

A travel request must be forwarded to the Contracting Officer's Representative in accordance with the contract terms for required travel. All travel must be approved by NOAA. Travel requests and justification shall be submitted and approved prior to travel. Approved contractor travel will be reimbursable and shall be in accordance with DoD Joint Travel Regulations where the contract terms are silent or are not applicable.

3.1.2 *Oil Analysis.* The Contractor shall work with the ships and the ashore analysis team to ensure that data such as Sample Point Identification (SPID) number and equipment nomenclature are correctly identified to the machinery to be sampled by the shipboard engineering crews. The Contractor shall ensure timely data transfer from the lab to the ashore database and from the ashore database to the ships. A Monthly Oil Data Analysis Report shall be submitted. Furthermore, the Contractor shall:

3.1.2.1 Update and deliver electronic equipment registration files to the current oil analysis lab.

3.1.2.2 Work with contracted oil analysis labs to troubleshoot sample delivery, tracking and analysis issues (for example, sample not received by the lab but SAMM indicates a sample was taken).

3.1.2.3 Analyze the used oil test results from the laboratory reports for the purpose of determining oil replacement or equipment maintenance and prepare feedback messages to vessels and supporting commands with the analyst results.

3.1.2.3.1 Review oil sample testing, test slate completion, laboratory sample analysis, ferrographic report ordering and content to ensure analysis accuracy.

3.1.2.4 Identify equipment groupings to facilitate creation of test variable alarms and then generate test variable alarms for all equipment where the data population is large enough.

3.1.2.5 Report used lube oil program results to management on a monthly basis.

3.1.2.6 Operate Shipboard Engineering Analysis System - Oil Analysis System (SEAS - OAS) for data maintenance, oil analysis, message preparation, reporting and secure file transfer.

3.1.2.7 Provide a fluid analysis contractor, to cover the cost of Lube oil analysis, Fuel analysis, Grease analysis, Transformer analysis, Sampling supplies and domestic US shipping.

3.1.3 *Vibration Analysis.* The Contractor shall use the Ashore Vibration Analysis Software (VAS) tools to allow the analysis of vibration monitoring data and to accurately predict machinery failures. The Contractor shall review recommendations made by the Shipboard Equipment Automated Diagnostic System (EADS) and provide recommendations based on the condition of the machine or the accuracy of the data collection. A Vibration Data Analysis Status Report shall be submitted monthly. Furthermore, the Contractor shall:

3.1.3.1 Review all serious and extreme (as defined by the SAMM database) EADS within three (3) working days. Any changes to the recommendations shall be communicated within SAMM to the affected ship's Chief Marine Engineer (CME), Port Engineer, and the Product Line Manager. Coordinate with the TPOC for contact information.

3.1.3.2 Upon request and approval by the Technical Point of Contact (TPOC) or Contract Officer's Representative (COR), the Contractor shall review requested EADS test results and vibration spectra within five (5) working days. Requests from NOAA shipboard personnel shall be forwarded to the TPOC or COR for concurrence prior to beginning the analysis. The Contractor shall report any changes to the recommendations to the affected ship's CME, Port Engineer, and the Product Line Manager.

3.1.3.3 Update the System Average Data, Machinery IDs (MIDs), Vibration Test Analysis Guide, and appropriate maintenance actions (for example, a maintenance action for misalignment may be to align the equipment). The Contractor shall identify any vibration software problems and provide support for developing and testing new software fixes and revising the system documentation. The Contractor shall provide a Monthly Vibration Data Analysis Report in an electronic format.

3.1.4 Develop the SAMM database for new NOAA Ships, install the database and program aboard new NOAA Ships, procure and deliver supporting SAMM system hardware, to train the ships force on how to use all the modules and hardware associated with the SAMM system, and to enroll new NOAA Ships SAMM maintenance plan into the ABS Preventative Maintenance Program (PM & CM).

3.1.5 Support the enrollment of NOAA vessels not yet participating in the ABS Preventative Maintenance Program (PMP), and ensure continued certification for those already enrolled by submitting Condition Monitoring (CM) data for critical equipment to ABS.

3.1.6 Provide a Monthly report of number on condition monitoring items (Oil and Vibration) due per ship and number completed per ship over the previous month.

3.1.7 Develop, maintain, and distribute Quick Reference Guides (QRG) for all SAMM versions active in NOAA's fleet.

3.1.8 Provide services to enter inventory from shore side warehouses (5) into the SAMM Spare Parts Module. Inventory data will be provided by NOAA personal located at each warehouse, the Contractor shall upload that data into the SAMM Spare Parts Module.

3.1.9 Port Engineering and Work Package Development Support

3.1.9.1 The contractor shall support NOAA Port Engineers in creating and maintaining each ship's Planned Maintenance Industrial Assistance (PMIA) index in the PMIA module in SAMM. At a minimum these indexes shall define the required industrial maintenance items for each ship to ensure that all regulatory body (ABS, USCG) and SMS requirements are met.

3.1.9.2 Provide SAMM PMIA development support. Develop the outline for PMIA actions. Provide a suggested list of SAMM Library items. Provide a relationship between PMIA Items and SAMM MCodes. The outline is intended to assist the Port Engineer during the development of Work Packages.

3.1.9.3 The Contractor shall provide data maintenance support for NOAA Cyclic Maintenance Plan and NOAA management. This support requires updating and management of reference data elements, voyage repair requests (VRR), Ship Alteration, and (PMIA).

3.1.9.4 The Contractor shall provide support for Port Engineers, such as properly mapping VRRs and PMIA items, in developing work packages, and availability execution.

3.1.10 Ship Inspection Program (SIP)

3.1.10.1 The Contractor shall provide an inspection team to conduct the material condition inspection of 3 NOAA Ships per year. Each Material Condition Assessment (MCA) shall consist of interviews and a multi-day ship check. Interviews with the Port Engineer, Officers, and Crew shall address known issues and issues currently being addressed. During the ship check, all available spaces are surveyed. Pictures of identified issues are taken to facilitate reporting and record-keeping. The rating scale should align with the ABS Grading Criteria for Hull Inspection Maintenance Program.

3.1.10.2 The contractor at the conclusion of the inspection and prior to departure shall provide an out brief with the Ship's Commanding Officer (CO), CME, and Port Engineer to go through all grade 5 & 6 findings.

3.1.10.3 The Contractor shall provide an MCA Summary Report including:

- 3.1.10.3.1 Summary of findings by type of issue
- 3.1.10.3.2 Summary of findings by type of space and recommendations for correction
- 3.1.10.3.3 Table of findings

3.1.10.4 The Contractor shall provide an excel spreadsheet of all ratings and findings and upload a Work Request in SAMM for every deficiency rated grade 3 or greater on the ABS Grading Scale discussed above including photos.

3.1.11 *Hardware Support*

3.1.11.1 The Contractor shall provide support and technical assistance to include installation, updates, testing, troubleshooting, for all existing and new engineering hardware and software including vibration data collectors, and associated support sensors/transducers. The Contractor shall provide for calibration, repair, and maintenance of all existing and new collectors, sensors, and hardware. The Contractor shall maintain a maintenance history log for all hardware and software to include the NOAA Personal Property Management System (PPMS) Sunflower applicable Property Identification Number (PIN).

3.1.11.2 The Contractor shall complete DD 1149 when transferring hardware to and from NOAA vessels.

3.2 SAMM Consolidated Databases & Replication Support

3.2.1 The Contractor shall provide support for the NOAA Shore Host Server(s) and for approximately 15 ships running SAMM to include the following:

3.2.1.1 Monitor and provide technical support for the data replication between the NOAA Shore Host Server(s) and installed ship SAMM databases. Technical support shall be available 24 hours a day, 7 days a week via email, phone, and on-site visit if not resolved remotely. Troubleshoot and Report problems in deliverable 3.1 Replication Status Report.

3.2.1.2 Provide a bi-weekly Replication Status Report for the current status of replication activity.

3.2.1.3 Failure to meet these service levels may result in remedies or deductions as specified in the contract. All services shall be delivered at the Firm-Fixed-Price agreed upon and shall not be subject to adjustment based on the level of effort required.

3.2.2 Provide a Monthly Summary Report identifying changes to the database for each ship to the NOAA TPOC and COR by the 10th of each month for changes made in the previous month.

3.3 Feedback Support

3.3.1 *Data Management.* The Contractor shall manage the (SAMM) data which is installed onboard NOAA vessels and ashore installations used in support of NOAA vessels by ashore personnel. The data management shall consist of the following:

3.3.1.1 Performing initial preventive maintenance feedback request reviews for analyzing the course of action to meet the request.

3.3.1.2 Conducting research of the issues for each feedback.

3.3.1.3 Submitting feedback proposals to the NOAA TPOC for the specific ship class that the feedback requests.

3.3.1.4 Conducting research of related PMIA work item specifications when related to equipment and/or maintenance changes. The contractor shall be responsible to review and propose changes to PMIA work item indexes when changes to PMIA and/or related equipment occurs.

3.3.1.5 Updating the SAMM database based on approved proposals.

3.3.1.6 Feedback Turnaround Time. The Contractor is expected to close out all feedback within sixty (60) days from the feedback origination date.

4 Program Management Support:

4.1 *Project Management Plan (PMP):* Develop a PMP, designate an Assistant Program Manager – APM NOAA, and submit to the TPOC within ten (30) business days of task award. The PMP should include all milestones, timeframes, resources (people), dependencies, deployment, communication matrix and Quality Assurance (QA) plan. The QA plan shall consist of procedures, metrics and remedial actions. The milestones in the PMP will be summarized in a baseline Gantt chart. All schedule changes shall be affected to the baseline Gantt chart.

4.1.1 *Trip Reports.* The Contractor shall provide Trip Reports to the TPOC for each ship visit within 14 days of the ship visit.

4.1.2 *Monthly Management Report.* The Contractor shall provide a Monthly Management Report to the TPOC that outlines the progress for each of the support requirements and brief this report no later than Tuesday of the second full week of each month to discuss any issues or recommended communications to increase participation in the SAMM program. The report shall provide the level of work performed during the month, including, the type of work and ships supported, ship visits, help calls received and corrected. The estimates for the percentage for each of the above activities/projects for the overall task are provided. Any additional work required or extensions will require submission of a Condition Found Report.

4.2 Government Provided Resources.

4.2.1 The Government will provide the following

4.2.1.1 Access to the NOAA SAMM Shore Host Servers located in the Cloud.

4.2.1.2 The SAMM server and workstation computers used on NOAA ships running SAMM.

5 DELIVERABLES:

No.	Deliverable:	Section	Format	FREQ
5.1	SAMM System Manuals	3.1.6	PDF	Upon installation and when updated
5.2	Monthly Oil Analysis Status Report	3.1.2	PDF/Excel	MONTHLY
5.3	Monthly Vibration Data Analysis Status Report	3.1.3	PDF/Excel	MONTHLY
5.4	Hardware & Software Maintenance History Log	3.1.10	PDF/Excel	MONTHLY
5.5	Replication Status Reports	3.2.1.2	PDF/Excel/Email	Bi-Weekly
5.6	Summary Report	3.2.2	PDF/Word	MONTHLY
5.7	Management Reports and Presentation	4.1.2	PDF/Excel	MONTHLY
5.8	Feedback Activity Report	3.3	PDF/Excel	MONTHLY

5.9	Support Activity Reports	4.1.2	PDF/Word/Excel	MONTHLY
5.10	Trip Report	4.1.1	Word	14 days after trip end
5.11	Condition Monitoring Report	3.1.5	Excel	MONTHLY